Elian Echavarria Avila

(929) 303-8649 | eechavarria.2022@gmail.com | linkedin.com/in/ElianEchavarria | github.com/ElianEchavarria | New York, NY

Education

Lehman College, CUNY — Bronx, NY
Bachelor of Science in Computer Science — *Jan 2026 – Expected Dec 2027*

Experience

Blackstone LaunchPad | **ServiceNetZero** — New York, NY
Business Development Intern — *June 2026 – Aug 2026*

- Architected v3 of the LabelZ platform in **TypeScript**, **Python**, and **Vite** (AI-assisted with **Claude** and **IBM Bob**, **Base44** for user profiles), taking the company from idea to MVP and giving ServiceNetZero its first sellable product
- Rebuilt the company landing page with **SEO**, **AIO/GEO** (LLM visibility), and **structured-data schema**, driving an immediate **22%** increase in LLM visibility and a spike in **Google Analytics** search performance
- Built a dashboard visualizing quantitative and qualitative non-LLM market data, improving research efficiency by over **60%**
- Sourced and completed the NSF I-Corps regional bootcamp, securing **$3,000** in customer-discovery funding; ran daily **Agile** standups and translated non-technical stakeholder goals into technical execution

Cambio Labs — Remote
Full-Stack Engineer Intern — *Feb 2026 – May 2026*

- Built a **23-step React/TypeScript** onboarding tour taking instructors from course creation to publishing on **10+** admin pages
- Built a custom spotlight/tooltip overlay from scratch with async-element support, viewport-aware placement, and dragging
- Centralized tour state in a **React Context** across nested routes, tracking progress and IDs so Skip/Back/Resume route correctly
- Made steps action-driven, advancing only when users complete the real task (e.g., creating a course or uploading media)

NYC Tech Talent Pipeline — New York, NY
Software Engineering Trainee — *June 2025 – Aug 2025*

- Built two full-stack apps on four-person teams (**JavaScript**, **React**, **Node.js**, **Express**, **SQL**): **Poll Maker** for creating and tracking interactive polls, and an **AI study assistant** that generates quizzes, tracks study habits, and calculates grades

Projects

AlgoStrike | *Next.js, Node.js/Express, PostgreSQL, Socket.IO, Docker* | algostrike.dev — 2026

- Architected and shipped a full-stack, real-time 1v1 competitive coding platform solo with **Socket.IO**, where players are matched live, solve the same problem head-to-head, and race to win; deployed to production over HTTPS
- Built a secure multi-language code-execution engine running untrusted **Python**, **JavaScript**, and **Java** submissions in isolated, network-disabled **Docker** containers with strict CPU/memory/**5-second** limits and an automated grader for **50+** problems
- Owned end-to-end deployment: provisioned an **Ubuntu VPS** with Docker, configured **nginx** as a reverse proxy with **Let's Encrypt TLS** and **WebSocket** support, ran Node under **pm2**, hosted the frontend on **Vercel**, and connected managed **Postgres**

Lyriq – Real-Time Synced Lyrics for Spotify | *Next.js, TypeScript, React, PostgreSQL, Prisma, Auth.js, Vercel* — 2026

- Built a full-stack **Next.js** app syncing live **Spotify API** playback to word-by-word karaoke lyrics on **Neon Postgres**
- Engineered a fault-tolerant lyrics pipeline that falls back across **3** sources (LRCLIB exact match, LRCLIB search, Genius scraping) with per-source timeouts and caching, parsing LRC timestamps and interpolating word timing when only line timing exists
- Designed a client-side playback clock extrapolating song position between polls via `requestAnimationFrame` with user-tunable latency compensation; adaptive polling (3s playing, 10s paused, off in background tabs) cuts redundant calls
- Implemented **OAuth 2.0** sign-in via **Auth.js** with automatic token refresh, plus a **Prisma/PostgreSQL REST API** for saved lyric "moments" with input validation, cursor-based pagination, and per-user data isolation
- Shipped in-browser video export that renders karaoke animations to **HTML Canvas** and records native MP4 via the **MediaRecorder API**, replacing a heavier planned WebAssembly encoder

Discord Stock Trading Game | *Python, asyncio, SQLite, yfinance, pytest* — 2026

- Built an async Discord bot (**discord.py 2.x**) running a paper-trading game on live market data, with persistent portfolios over a normalized **SQLite** schema (users, holdings, trade ledger), P&L tracking, and a scheduled weekly leaderboard
- Implemented weighted-average cost-basis accounting to compute realized and unrealized P&L across partial buys and sells
- Cut multi-symbol quote latency ∼**5x** by replacing sequential awaits with `asyncio.gather` and adding a 60s TTL cache that collapses repeated ticker lookups into one API call; kept all I/O off the event loop via `asyncio.to_thread` and **aiosqlite**
- Wrote **27 pytest** tests with mocked data and temp databases; mutation testing exposed a test that passed against broken math

Leadership & Community

ColorStack Chapter, Computer Science Club | *Technical Team Member* — Oct 2025 – Dec 2025

- Led LeetCode/DSA workshops and mentored peers in interview prep; open-source contributor in Hacktoberfest 2025

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS | **Spoken**: English, Spanish (fluent)
Frameworks & Libraries: React, Next.js, Node.js, Express, Prisma, Auth.js, Socket.IO, Tailwind CSS, Vite, discord.py, pytest
Databases: PostgreSQL, SQLite, Sequelize (ORM), schema design, data visualization
Infrastructure & Tools: Docker, nginx (reverse proxy, TLS), pm2, Vercel, Git, GitHub, Bash, npm, Postman, Figma, Claude
Concepts: REST APIs, OAuth 2.0, WebSockets, OpenAPI, HTTP, Agile, SEO/AIO/GEO, responsive design